

The Power of Many: Multi-User Collaborative Indoor Localization for Boosting Standalone User-Based Systems in Different Scenarios

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BIOGRAPHY

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ABSTRACT

In today's smartphone-driven era, our lives revolve around smartphones, resulting in high smartphone densities in various contexts. Furthermore, modern smartphones can exchange inter-user measurements via different sources, such as WiFi and BLE. In turn, the concept of online multi-user collaboration has recently emerged as a promising solution to improve the performance of standalone user-based indoor positioning systems (SU-IPSS). On this basis, this research proposes a real-time multi-user collaborative indoor positioning (RT-MUCIP) scheme. This scheme aims at boosting the indoor positioning performance of users with unavailable position information or low position confidence. The procedures of the proposed scheme can be summarized as follows: First, it checks the density and position confidence of the surrounding users. Users with high position confidence are identified. Subsequently, the proposed scheme adapts to the changing density of users in the following manner: In scenarios with sparse users, a nearby user with high confidence is explored and exploited to boost a near-neighbor with low position confidence. In areas with dense users, the weight of surrounding users' positions and inter-user measurements are determined, and the RT-MUCIP solution is estimated using a weighted non-linear least squares algorithm. Additionally, inspired by wireless sensor networks, RT-MUCIP scheme proposed method to upgrade users observed in static mode with high positioning confidence to act as temporary anchor points. As a result of this upgrade, anchor node density increases, and overall positioning performance can be improved. To evaluate the performance of the proposed scheme, several tests were conducted in three scenarios. In light of the tests results, we can conclude that the proposed

collaborative localization scheme can improve the localization accuracy of collaborated users without the need to use external resources.

I. INTRODUCTION

Today, location awareness and location-based services (LBS) are required everywhere by service providers and customers (Zhang & Hsu, 2021). However, due to signal attenuation and blocking caused by obstructions such as walls and roofs, global positioning systems (GNSSs) perform significantly worse indoors than outdoors (Wen et al., 2021). Even so, precise indoor positioning is necessary for a wide range of services of services, including commercial, healthcare, and safety services (Mansour & Chen, 2022). Alternative technologies have consequently arisen in an effort to address the localization challenges in indoor environments. The common alternatives technologies include Wi-Fi (Mansour et al., 2023), Bluetooth low energy (BLE) (Luo et al., 2021), ultra-wideband (UWB) (Zhou et al., 2022), infrared, map matching (Wang et al., 2023), and sensor fusion approaches.

These alternative technologies have primarily been employed over the past two decades to develop a kind of standalone user-based indoor positioning systems (SU-IPs), where a user's location is determined solely by his or her own observations including, ambient wireless signals, inertial measurements, or the integration of both. SU-IPs could achieve an accurate solution in specific scenarios; however, their accuracy and reliability are prone to variability and degradation under different scenarios due to 1) the complexity and diversity of indoor environments; 2) the density and distribution of wireless anchor nodes; 3) the signal propagation problems in indoor areas; and 4) the low-grade of smartphone's sensors (Suzuki, 2023).

Recently, real-time multi-user collaborative indoor positioning (RT-MUCIP) has emerged as a promising approach to boost SU-IPs and overcome their limitations. The emergence of RT- MUCIP is driven by several factors, including the high density of smartphones in various indoor contexts provides ample opportunities for cooperation. Such density is due to the ubiquitous presence of smartphones in our daily lives since they provide end-users with unlimited access to a vast array of information, services, and communication. More importantly, modern smartphones are equipped with communication capabilities that enable the exchange of measurements between devices, thereby facilitating collaborative localization implementation. On this basis, this paper proposes RT-MUCIP scheme that aims at boosting the indoor positioning performance of users with unavailable position information or low position confidence. The key *contributions* of this study can be summarized as follows: Introducing method to leverage the real-time multi-user collaboration to improve the positioning performance of SU-IPs. The merits of this scheme are as follow: 1) the scheme proposes adaptive algorithm based on the position confidence of the surrounding users and the user density; 2) it can be exploited for both stationary and walking modes; and 3) to improve the anchor node density and distribution, the scheme proposes a method for upgrading stationary users with qualified criteria to act as temporary anchor nodes.

The remainder of this paper is organized as follows. Section II reviews the existing literature related to leveraging multi-user collaboration in smartphone-IPs. Section III discusses the proposed methodology. The system development, experiments, results and evaluations are presented in section IV, followed by conclusions in section V. Table 1 lists the abbreviations used in this paper.

Table 1 List of Abbreviations

Abbreviation	Definition
AN	Anchor Node
AP	Access Point
BLE	Bluetooth-Low-Energy
GNSS	Global Navigation Satellite System
IPS	Indoor Positioning System
LBS	Location-Based Services
LoS	Line-of-Sight
MDS	Multi-Dimensional Scaling
NLoS	Non-Line-of-Sight
PF	Particle Filter
P2P	Peer-to-Peer
PDR	Pedestrian Dead Reckoning
PS	Peripheral Simulator
RP	Reference Point
RT- MUCIP	Real-Time Multi-User Collaborative Indoor Positioning

RSS	Received Signal Strength
SU-IPS	Standalone User-Based Indoor Positioning System
UWB	Ultra-Wideband
UUID	Universally Unique Identifier
WSN	Wireless Sensor Network

II. RELATED WORK

This section reviews the existing pedestrian indoor positioning studies that leveraged the real-time multi-user collaboration to improve the performance of SU-IPSs. The existing studies can be broadly categorized based on the need for exchanging inter-user information into implicit and explicit collaborative schemes. Implicit schemes exploit the latent signaling relationships between two or more users to refine the positions estimated individually without the need of exchanging inter-user communication or transmissions. OCLoc scheme developed by [Zhou and Wang \(2020\)](#) is the only scheme that compared the observed signal vectors of collaborated users with the RPs vectors to ensure high credibility of fingerprinting solution. Although OCLoc could be effective in scenarios where the users' devices are similar with a low offset of signal reception, OCLoc has practical limitations in real-world scenarios with heterogeneous devices ([Wang et al., 2016](#)).

Instead of OCLoc, most proposed collaborative schemes explicitly exchange inter-user measurements (e.g., spatial proximity or physical range) among cooperating devices. These measurements can be utilized to constrain and refine the positions estimated individually from their underlying localization systems. Schemes-based proximity sensing utilizes proximity sensors or wireless signals to decide whether two devices are nearby or encounter each other. Commonly, proximity was exploited to improve PDR or Wi-Fi fingerprinting techniques. [Chan et al. \(2006\)](#) proposed the early collaborative indoor positioning scheme where the proximity was sensed by ZigBee radios to improve the fingerprinting performance. The other proposed proximity sensing-based collaborative schemes aimed to enhance PDR performance ([Chen et al., 2017](#); [Higuchi et al., 2014](#); [Jun et al., 2013](#)). For example, the Social-Loc system developed by [Jun et al. \(2013\)](#) detected encounter and non-encounter events to update the weights of the particles in a fusion scheme based on particle filter (PF). The EcoLoc scheme proposed by [Chen et al. \(2017\)](#) exchanged the trajectories of encountering users and merged them to calibrate a tracking algorithm using a conditional random forest approach. People-centric navigation (PCN) scheme developed by [Higuchi et al. \(2014\)](#) utilized Bluetooth RSS for proximity sensing to find a user in the crowd.

The inter-user distance obtained by inter-user ranging techniques (e.g., acoustic and radio frequency (RF) signals) has been utilized to constrain users' possible locations and enhance localization SU-IPSs. Similar to collaboration based on proximity sensing in PDR, inter-user distances has been used to adjust the particle weights in the PF fusion in different studies, such as ([Liu et al., 2020](#); [Noh et al., 2016](#); [Qiu & Tseng, 2016](#)). Moreover, inter-user distance has been exploited for Wi-Fi fingerprinting to refine the best candidate reference points (RPs) estimated by underlaying/standalone-user fingerprint matching ([Chen et al., 2016](#); [Jing et al., 2016](#); [Liu et al., 2013](#); [Nandakumar et al., 2012](#); [Sakr et al., 2020](#); [Seco & Jimenez, 2018](#); [Taniuchi et al., 2014](#)). Using acoustic ranging, [Liu et al. \(2013\)](#) proposed constructing a graph of users with an edge representing the physical distance between each user. To refine fingerprinting locations, the users' graph is superimposed over a fingerprint map, rotated, and translated. They, however, assumed an accurate distance and imposed a too stringent distance restraint on their user graph. A Bayesian network framework and a Bayesian graphical model are used by Centaur scheme developed by [Nandakumar et al. \(2012\)](#) to identify probable locations of collaborated users. [Chen et al. \(2016\)](#) proposed a centralized cooperative multidimensional scaling (MDS) that utilized an offline database of Wi-Fi signatures and inter-user distances. Multi-user indoor environments were targeted and effectively improved positioning precision to cope with errors of outdated databases. [Taniuchi et al. \(2014\)](#) proposed a collaborative localization scheme based on a spring model where the inter-user distance was considered as a kind of force that attracts or repels users. [Seco and Jimenez \(2018\)](#) fused the PDRs, map data, and inter-user distances estimated from exchanging RSS measurements between users using active RFID mobile tags. LocSpeck, developed by [Sakr et al. \(2020\)](#), introduced a distributed positioning algorithm based on PF and bi-directional UWB communication networks.

Reviewing the related studies reveals that the existing studies propose different collaborative localization methods to boost the underlay techniques of SU-IPS. However, our research goes beyond existing studies by leveraging multi-user collaboration in diverse indoor scenarios, adapting the collaborative localization approach to accommodate varying user densities, and identifying stationary users with high position confidence to serve as fixed anchor nodes, thereby augmenting the anchor node density.

III. METHODOLOGY

1. Outlines of the Proposed Approach

This subsection provides the outlines of the proposed approach. First, the motion mode of each node—specifically, whether it was in a static or kinematic mode—was checked when a group of users surrounded a user (the target node). Notably, the motion mode impacts the user's role in the collaborative scheme, position and confidence estimation. A user position was estimated individually, following that the confidence of the estimated position was determined; the readers are referred to subsection 2 for more details. From the exchanged inter-user measurements, inter-user distances were calculated, as well as the confidence in the distance estimation; see subsection 3 for more information. Afterwards, the proposed scheme checked the density and position confidence of the surrounding users. As a consequence, the proposed scheme adapts to changing user density in the following way. When there are spare users nearby, a user with high confidence is explored and exploited to boost a neighbor with low confidence in their position. In areas with dense users, the inter-user distances were computed from the exchanged inter-user measurements, and each estimated distance's confidence was derived. Each surrounded node contributed to the collaborative positioning estimation based on its position and inter-user distance confidence. Then the collaborative position was estimated using a weighted non-linear least squares algorithm that considers the weight of node positions and inter-user distances. Furthermore, inspired by wireless sensor networks, the proposed scheme also upgrades users observed in static mode with high positioning confidence to act as temporary anchor points. This upgrade aims to increase anchor node density to improve overall positioning performance. Figure 1 represents the flowchart of the proposed online collaborative localization approach.

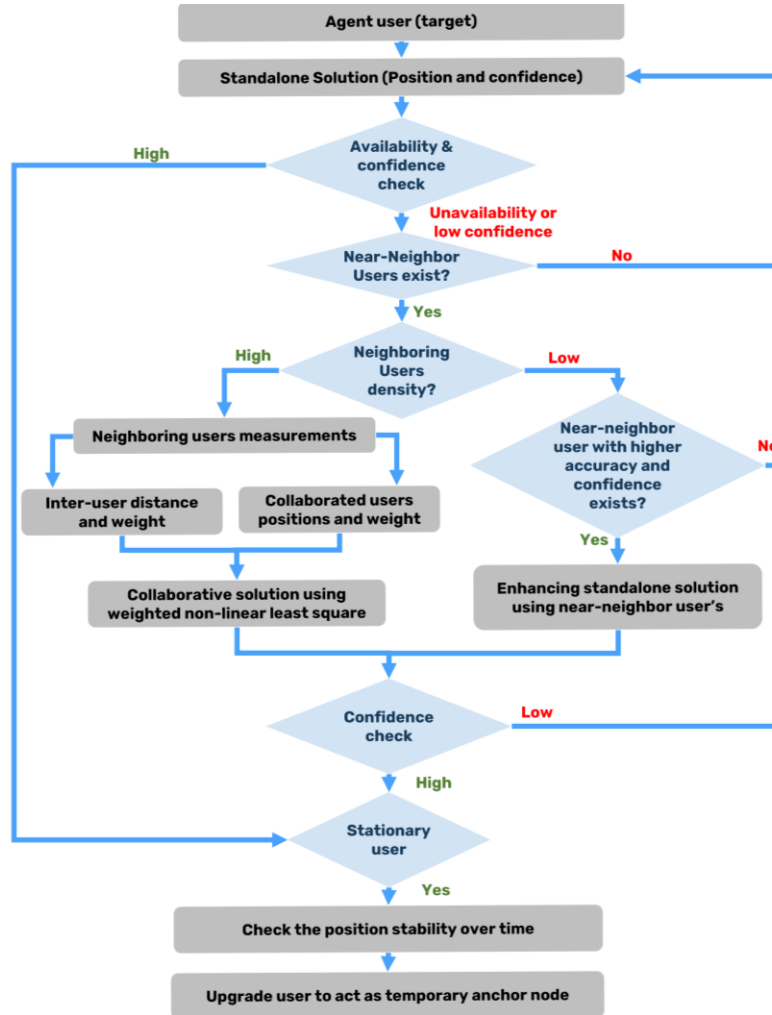


FIGURE 1 Flowchart of the proposed scheme for multi-user collaborative indoor positioning

2. Standalone User-Based Indoor Positioning

This subsection presents the methodology of positioning a standalone user and the confidence in its positioning estimate in an indoor environment.

a) Stationary Mode

Position Estimation: In static mode, time-series of Wi-Fi RSS data can be employed to compute a final position of a static node, either by computing the average over a set of estimated fingerprinting positions or smoothing the measurements over time by different methods (e.g., averaging) to compute the current position. Generally, at epoch t , a smartphone can receive an RSS vector from surrounding APs and retrieve the radio map database of the indoor area. As long as a user was in a static mode, the current RSS observed for a specific AP was smoothed using the previous set of RSS data through moving averages to smooth out short-term fluctuations and preserve long-term trends. Comparing the smoothed online RSS vector with the offline reference points (RPs) enables estimating the current location $\mathbf{x}_{i, f_{ps}}^t$ through fingerprinting-based methods in a non-collaborative localization approach. The online position was estimated using fingerprinting approach based on K-Weighted Nearest Neighboring (KWNN). Our approach relies on a Euclidean distance summation between offline RPs and online Wi-Fi RSS signatures to determine the best K neighboring RPs. The online position was estimated as follows:

$$\mathbf{x}_{i, f_{ps}}^t = (x_{f_{ps}}, y_{f_{ps}}) = \left(\sum_{i=1}^K (x_i^{RP} \cdot w_i), \sum_{i=1}^K (y_i^{RP} \cdot w_i) \right) \quad (1)$$

where (x_i^{RP}, y_i^{RP}) is the location of a candidate RP $\forall_{i \in \{1 \dots K\}}$, K is the predefined number of RPs to estimate the online fingerprinting solution, and w_i is the contribution weight of each selected RP.

Confidence of the Estimated Position: after estimating a user position, a confidence level of this estimation is required to determine the extent to which this solution can be trusted. This level should indicate whether the estimated location is close to its actual location at a high or low probability. Based on this level, the collaborative localization algorithm can decide whether the estimated position of a user qualifies to contribute to improve the location estimation of the surrounding users or needs to be improved by surrounding users. In stationary mode, the signal strength variation can be used to determine the positioning confidence level. This is attributed to the following factors: Fingerprinting-based positioning methods face challenges due to the susceptibility of position estimates to errors caused by variations in received signal strength (RSS). These variations stem from multiple factors, including small-scale effects and multipath interference, which can cause significant fluctuations in signal strength when a mobile device moves within the wavelength range (e.g., 12.5 cm for Wi-Fi). Consequently, even in static scenarios during both offline and online phases, minor movements can significantly affect the measured RSS values, undermining the accuracy of the positioning estimates that rely on implied probabilities or weighted values derived from these RSS samples. The movements in the surroundings (e.g., a person walking around or a door opening or closing) cause radio signals to travel differently, thereby causing the RSS to fluctuate. The probabilities of selecting the actual nearest RP are relatively minor if there is a considerable variation between different RSS samples, so it is more likely that a remote RP would be selected. Hence, an increase in variance also increases the risk of significant position errors and reduces the confidence level of the estimated position. Therefore, the RSS variation was used to reflect the confidence level for the estimated position. The average variance of the RSS variation was estimated as follows: 1) for each AP that is part of the RSS samples, the mean RSS (P_{AP}^m) was determined; 2) based on the P_{AP}^m , all RSS samples available for this given AP were subtracted; 3) the RSS standard deviation for each AP was then calculated, and 4) the overall average RSS standard deviation from all APs was calculated. As proposed by (Lemelson et al., 2009), the following formula donates the average standard deviation estimate for all APs:

$$\vartheta = \frac{\sum_{AP=1}^N \left(\sqrt{\frac{\sum_{i=1}^M ((P_{AP}^m - P_{AP}^i)^2)}{M}} \right)}{N} \quad (2)$$

where N is the number of observed APs, and M is the number of samples per AP. Accordingly, the confidence of the estimated position was calculated based on the average standard deviation of RSS as follows:

$$CL_{pos}^{rss} = e^{-\vartheta} \quad (3)$$

The RSS variation directly indicates the stability of measurements and the impact of the surrounding environment on the measurements while estimating a user position. However, the impact of this variation on the estimation process could be changed from one scenario to another due to different factors, such as the density of the APs, geometric distribution of the existing APs concerning a user location, and the maximum strengths observed from these APs. Therefore, determining the variance of the estimated locations while a user is in a static mode can be feasible to obtain another indication of the confidence level of the final estimated position. Thus, the confidence score of the final (averaged) estimated position was also estimated as a function of the standard deviation of the estimated positions. Accordingly, the confidence of the estimated position was calculated as follows:

$$CL_{pos}^p = e^{-\sigma_i} \quad (4)$$

where σ_i is the standard deviation of estimated positions for the i^{th} static node. Two predefined thresholds were set to define the confidence level for CL_{pos}^{rss} and CL_{pos}^p . A user with an estimated level lower than one of the two thresholds was considered a user with low position confidence.

b) Kinematic Mode

Position Estimation: The proposed method employs PDR as the chosen approach for establishing the fusion model and updating the user's location. This decision stems from the recognition that step detection is better suited to accommodate the inherent limitations in accuracy associated with low-cost accelerometers commonly found in smartphones. Due to the inherent nonlinearity of the PDR model, the fusion of PDR and Wi-Fi fingerprinting solutions was accomplished using the extended Kalman filter (Julier & Uhlmann, 2004). Utilizing the PDR method, which can be explored further in (Mansour et al., 2021), the user's position (x_k, y_k) was promptly updated upon the detection of a new step, denoted as k . The position and heading updates based on the PDR method can be represented by the following formula:

$$\begin{cases} x_k = x_{k-1} + (1 + s_k) sl_k \sin(h_k) + \sigma_x \\ y_k = y_{k-1} + (1 + s_k) sl_k \cos(h_k) + \sigma_y \\ s_k = s_{k-1} + \sigma_s \\ h_k = h_{k-1} + \Delta h_k + \sigma_h \end{cases} \quad (5)$$

where sl_k and s_k represent the moved distance and the distance correction, respectively. h_k and Δh_k donate the heading and heading change, respectively. $\sigma_x, \sigma_y, \sigma_s$ and σ_h are the east, north, distance, and heading noises, respectively. Accordingly, the state vector can be defined as $\mathbf{x} = [\delta_x \ \delta_y \ \delta_{sl} \ \delta_h]^T$, where δ is the error of each parameter. To linearize the transition model, a partial derivative was applied. Building upon this, the state equation can be described as follows: $\mathbf{x}_t = \mathbf{A}_{t-1|t} \mathbf{x}_{t-1} + \mathbf{B} \boldsymbol{\omega}_t$, where $\mathbf{A}_{t-1|t}$ indicates the state transition matrix from the epoch $t-1$ to t , $\mathbf{B}$ is the system noise matrix, and $\boldsymbol{\omega}_t$ is the system noise vector with covariance matrix $\mathbf{Q}$: $\boldsymbol{\omega}_{t-1} \sim N(0, \mathbf{Q}_{t-1})$. The transition matrix was derived by linearizing the transition model, resulting in the following formulation:

$$\mathbf{A}_{t-1|t} = \begin{bmatrix} 1 & 0 & sl_k \sin(h_k) & (1+s) sl_k \cos(h_k) \\ 0 & 1 & sl_k \cos(h_k) & (-1-s) sl_k \sin(h_k) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

The observations model can be expressed as: $\mathbf{z}_t = \mathbf{C}_t \mathbf{x}_t + \mathbf{v}_t$, where $\mathbf{C}_t$ is the matrix used to estimate the predicted measurement $\mathbf{z}_t$ from the predicted state $\mathbf{x}_t$, and $\mathbf{v}_t$ is the observation noises which is assumed zero-mean with the covariance matrix $\mathbf{R}$: $\mathbf{v}_t \sim N(0, \mathbf{R}_t)$. The localization results obtained from PDR and fingerprinting were incorporated into the observation vector as $\mathbf{z}_t = [\Delta x \ \Delta y \ \Delta s \ \Delta h]^T$, where Δ is the bias in each parameter. The transition matrix $\mathbf{C}$ was defined as $diag(c_1, c_2, c_3, c_4)$ and covariance matrix $\mathbf{R}$ was donated as $diag(r_1, r_2, r_3, r_4)$. The parameters of EKF were updated based on the ambient environment and the available measurement. The implemented procedures of EKF can be summarized as follows: 1) in the initialization stage ($t = 0$), the initial state and covariance were initialized as follows:

$$\begin{aligned}\hat{X}_o &= A_o X_o \\ P_o &= P_o\end{aligned}\quad (7)$$

where X_o , A_o , and P_o are the initial state, transition, and covariance matrix; 2) After that, at the computation stage ($t > 0$), the steps were performed: a) the state estimate was predicted the, as follows:

$$\hat{X}_t^- = A_{t-1} \hat{X}_{t-1} \quad (8)$$

where $\hat{X}_t^-$ and $\hat{X}_{t-1}^-$ are the state predication matrix for the current epoch and estimated state for the previous epoch. The covariance was also predicted, as follows:

$$P_t^- = A_{t,t-1} P_{t-1} A_{t,t-1}^T + Q_{t-1} \quad (9)$$

where P_t^- and P_{t-1} are the covariance predication matrix for the current epoch and estimated covariance for the previous epoch; b) the Kalman gain was estimated, as follows:

$$G_t = P_t^- C_t^T [C_t P_t^- C_t^T + R_t]^{-1} \quad (10)$$

Then, the state estimate and covariance were updated, as follows:

$$\begin{aligned}\hat{X}_t &= \hat{X}_t^- + G_t [Z_t - C_t \hat{X}_t^-] \\ P_t &= (I - G_t C_t) P_t^-\end{aligned}\quad (11)$$

Finally, the integrated position of the user was estimated as follows:

$$\begin{cases} x_t = x_{t-1} + \delta_x + (1 + s + \delta_s) sl_k \sin(h_k + \delta_h) \\ y_t = y_{t-1} + \delta_y + (1 + s + \delta_s) sl_k \cos(h_k + \delta_h) \end{cases} \quad (12)$$

where δ_x , δ_y , δ_h and δ_s are the error for easting, northing, heading and distance correction, respectively.

Confidence Level of the Fused Position: the hypotheses of using RSS stability over time and the variance of the estimated positions as confidence indicators are feasible for a stationary device. Contrarily, these hypotheses cannot be used in kinematic mode since the RSS and user position are changed over time. Therefore, in this case, distances between the best selected nearest neighbors fingerprinting algorithms could be utilized to infer a confidence level of the estimated fingerprinting position. Selecting adjacent K-RPs should often exhibit highly overlapping RSS properties that can cover the range of online RSS variation. The expected position error can be estimated from the K best fingerprints returned from a fingerprinting algorithm. From the K best estimates returned from a fingerprinting algorithm, an expected error estimate can be deduced by calculating the average weighted distance (w_d) between the K best fingerprints, as follows.

$$w_d = \frac{\sum_{i=1}^K (w_i \cdot \sqrt{(x_{fps} - x_i)^2 + (y_{fps} - y_i)^2})}{\sum_{i=1}^K (w_i)} \quad (13)$$

where (x_i, y_i) is the location of the i th RP, and w_i is its estimated weight. K is the number of selected best nearest RPs. In the fusion solution-based EKF, the fingerprinting solution was utilized to update the inertial position, and the covariance matrix was updated using w_d . At epoch t , the updated covariance (P_t) was estimated based on Equation (11) of the new estimate was determined. Then, the confidence of the fused position was calculated as follows:

$$CL_{pos}^p = e^{-P_t} \quad (14)$$

On this basis, for user i in a kinematic mode, the estimated position and its confidence score were obtained as $(\langle x_t^i | CL_{pos,t}^i \rangle)$.

3. Inter-User Distance and Confidence Estimation

a) Distance Estimation

The proposed scheme acquires the inter-user distances through the BLE RSS measurements. The well-known path-loss (PL) model was utilized to convert the RSS to distance. The PL model can be defined as:

$$P_d = \lambda_t - 10\alpha \log_{10} d + v \quad (15)$$

where λ_t is the RSS at a distance 1m, P_d is the RSS at distance d in dBm, and α is the PL exponent due to the signal attenuation. v is a Gaussian random variable, and $v \sim \mathcal{N}(0, \sigma_v^2)$ with standard deviation σ_v . In the proposed cooperative localization scenario, agent user $\mathbf{x}_t$ ($t = N_a + 1, N_a + 2, \dots, N$) was considered as the target node (i.e., a user whose position should be estimated). The signal received at $\mathbf{x}_t$ from surrounding node $\mathbf{x}_j$ ($j = 1, 2, \dots, N$) is denoted by P_{tj} (i.e., BLE RSS between the target user and a surrounding user). Based on the PL model, the measured P_{tj} was converted to distance as:

$$\hat{d}_{tj} = 10^{\frac{\lambda_t - P_{tj}}{10\alpha}} \quad (16)$$

b) Confidence of Distance Estimation

After estimating the inter-user distance $\hat{d}_{tj}$, estimating the confidence of this distance is crucial for weighting the contribution of each node and ensuring robust collaborative localization. RSS may attenuate significantly in indoor environments when a barrier such as a wall or moving people exists. Electromagnetic interference may also weaken the RSS. In addition, RSS decreases as the distance between two nodes becomes long. Therefore, it is difficult to determine whether the low RSS results from long distances, obstacles, or interference. With high confidence, it can be supposed that the stronger RSS provides a more accurate distance estimation. Conversely, there is a low degree of confidence that weaker RSS results from long-distance. Consequently, the confidence of the estimated inter-user distance was calculated based on the estimated RSS as:

$$CL_{DP} = e^{\frac{P_{tj}}{\rho}} \quad (17)$$

where ρ is a constant equal to -110 dBm, utilized to describe the minimum of detectable RSS and normalize the RSS.

Within a short interval that the change in the two users' position ($\mathbf{x}_t$ and $\mathbf{x}_j$) can be neglected, theoretically, P_{tj} should be nearly equal P_{jt} and the absolute difference between them $\Delta P = |P_{tj} - P_{jt}|$ should be trivial. ΔP was also considered as a factor to compute the distance confidence. Obtaining low ΔP was considered as an indication of the low impact of distracting factors and homogeneity of the chips of two nodes, which in turn gives high confidence in the estimated distance. Accordingly, another indication for the confidence of the estimated inter-user distance was calculated based on ΔP as:

$$CL_{\Delta P} = e^{-\frac{|\Delta P|}{\Delta P_\tau}} \quad (18)$$

where ΔP_τ is a constant equal to 20 dBm, utilized to describe the acceptable absolute ΔP range and normalize the RSS difference. We considered the weight of CL_{DP} should be partially higher than $CL_{\Delta P}$. In case of CL_{DP} is supported by $CL_{\Delta P}$, the distance confidence should be more reliable than when CL_{DP} is the only exists, thus we summed CL_{DP} and $CL_{\Delta P}$ after weighting them. Consequently, the overall estimated confidence for estimated inter-user distance w_d was computed as:

$$w_d = 0.6 \cdot CL_{DP} + 0.4 \cdot CL_{\Delta P} \quad (19)$$

4. Real-Time Multi-User Collaborative Indoor Positioning (RT-MUCIP) Approach

a) Scenario of Low User Density

In real-world scenarios, there are instances where indoor areas exhibit low user density, but a nearby user may possess better position accuracy and confidence than the target user. This situation can arise in various scenarios, such as when a user (User A) is located

close to a window or Wi-Fi access point or transitions from an outdoor to an indoor area. This user with better position performance could meet another user (User B) in a deep indoor region with sparse access point density. In this case, user B could exhibit position unavailability or position estimated with low confidence. With the existence of inter-user RSS data, the inter user distance or how close two users are to each other can be determined. Based on the closeness range, the position of user B can be improved. The confidence of user B, in this case, can be estimated as: $CL_p^B = CL_p^A + f_{d_{AB}}$, where $f_{d_{AB}}$ is to derive the position confidence based on the proximity (d_{AB}) between two users. Implicit relationships can be derived from two users' RSS measurements observed from the same APs. These relationships can indicate the closeness or proximity between two users. However, such relationships require ensuring the homogeneity of collaborative node receivers' chips. Therefore, the first case was considered in this work, where inter-user RSS measurements are exchanged.

b) Scenarios of High User Density

With the growing prevalence of smartphones in our daily lives, it is common to find several indoor contexts with high smartphone density. In these scenarios, the position availability and confidence of the existing users could vary based on different environmental and device factors. Collaboration among multiple users can boost the position estimation of surrendering users. The following illustrates the proposed scheme to leverage collaboration among multiple users in high user density scenarios. At a given instant, let $\boldsymbol{\varphi} = [x, y]$ donates the location of a user (target user) and $\boldsymbol{\theta}_i = [x_i, y_i]$ provides the location of a collaborated node that can be detected by the target user, where $i \in \{1, \dots, M\}$. The following expression can give the distance between the i -th node and the target user:

$$\hat{d}_i = \|\boldsymbol{\varphi} - \boldsymbol{\theta}_i\| + \varepsilon_i + \delta_i = \sqrt{(x - x_i)^2 + (y - y_i)^2} + \varepsilon_i + \delta_i \quad (20)$$

where ε_i is the ranging error and δ_i is additive measurement noise. The target user can receive RSS measurements from M nodes and form an inter-distance vector donated by $\hat{\mathbf{d}} = [\hat{d}_1 \dots \hat{d}_M]^T$. Given M noisy measurements, the problem of least square localization is essential to obtain a solution from a set of nonlinear equations given by:

$$\mathbf{F}(\boldsymbol{\varphi}) = \begin{bmatrix} \sqrt{(x - x_1)^2 + (y - y_1)^2} \\ \vdots \\ \sqrt{(x - x_M)^2 + (y - y_M)^2} \end{bmatrix} \quad (21)$$

This nonlinear problem requires minimizing the cost function which can be expressed as follows:

$$E[\hat{\boldsymbol{\varphi}}] = [\mathbf{d} - \mathbf{F}(\hat{\boldsymbol{\varphi}})]^T [\mathbf{d} - \mathbf{F}(\hat{\boldsymbol{\varphi}})] \quad (22)$$

The nonlinear function can be linearized using the first-order Taylor expansion around $\boldsymbol{\varphi}_0$ and retaining the first two terms as $\mathbf{F}(\boldsymbol{\varphi}) \approx \mathbf{F}(\boldsymbol{\varphi}_0) + \mathbf{H}(\boldsymbol{\varphi} - \boldsymbol{\varphi}_0)$, where $\mathbf{H}$ is the Jacobian of $\mathbf{F}$ donated by:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \vdots & \vdots \\ \frac{\partial f_M}{\partial x} & \frac{\partial f_M}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{x - x_1}{\sqrt{(x - x_1)^2 + (y - y_1)^2}} & \frac{y - y_1}{\sqrt{(x - x_1)^2 + (y - y_1)^2}} \\ \vdots & \vdots \\ \frac{x - x_M}{\sqrt{(x - x_M)^2 + (y - y_M)^2}} & \frac{y - y_M}{\sqrt{(x - x_M)^2 + (y - y_M)^2}} \end{bmatrix} \quad (23)$$

The linearized least square solution can be donated by:

$$\hat{\boldsymbol{\varphi}} = \boldsymbol{\varphi}_0 + (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T [\mathbf{d} - \mathbf{F}(\boldsymbol{\varphi}_0)] \quad (24)$$

Localization performance can be improved by applying a weighted least square (WLS) adjustment method that considers the weight of node positions and inter-user measurements. Consequently, the weights of the inter-distance estimation: $\mathbf{w}^d = [w_1^d \dots w_M^d]^T$ and nodes positions: $\mathbf{w}^p = [CL_1^p \dots CL_M^p]^T$ were associated with the inter-user distance and node position vectors. The WLS algorithm solution is formed as the vector $\hat{\boldsymbol{\varphi}}$ that minimizes the cost function:

$$E_w[\hat{\varphi}] = [\mathbf{d} - \mathbf{F}(\hat{\varphi})]^T \mathbf{W} [\mathbf{d} - \mathbf{F}(\hat{\varphi})], \mathbf{W} = \text{diag}(w_1 \dots w_M) \quad (25)$$

where $\mathbf{W}$ is a diagonal matrix that combines the weights of inter-distance and position, and w_i is a combination of the inter-distance weight and position weight or formally given by:

$$w_i = \frac{1}{\left(1/w_d^i\right) + \left(1/w_p^i\right)} \quad (26)$$

The minimization of E_w converts the LS to:

$$\hat{\varphi} = \varphi_0 + (\mathbf{H}^T \mathbf{W} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{W} [\mathbf{d} - \mathbf{F}(\varphi_0)] \quad (27)$$

Then the variance of the positioning error was estimated from the error covariance donated as follows:

$$\sigma^2 = \text{trace}((\mathbf{H}^T \mathbf{W} \mathbf{H})^{-1}) \quad (28)$$

The singular value decomposition (SVD) was utilized to determine σ_x^2 and σ_y^2 of the estimated location error covariance.

c) Upgrading Agent User to Temporary Anchor Nodes (TAN)

Once a user has estimated its position and its position variance, it enrolls into the TAN nominee phase, where it could be upgraded to operate as an anchor node. The benefit of this upgrading is extending the availability and increasing the anchor nodes' density, especially for the areas with spare anchors. The position variance of a nominated node was first compared with a predefined threshold. If the node position had a variance less than this threshold, it was nominated as TAN. Then the user's walking state was observed over time. If the smartphone was in a static mode, the estimated positions over time were recorded to ensure higher stability. The variance of the estimated positions was then computed. While this variance was lower than a threshold, this static user was upgraded to TAN.

IV. SOFTWARE DEVELOPEMENT, EXPEREMINTS, RESULTS, AND EVALUATIONS

1. Software Development

The following steps were developed to enable conducting real-time collaborative localization in real-world scenarios: 1) To enable emitting signals and exchange of inter-user measurements: BLE Peripheral Simulator was included in the developed JAVA code for real-time operations. The functionality of the BLE simulator is to broadcast BLE RSS from a smartphone with a unique ID (Major, Minor, and UUID). With broadcasting, a smartphone can receive BLE RSS from the surrounding users. A JAVA code was developed to demonstrate these two functions; 2) The observed Wi-Fi and BLE RSS data, user status, estimated position, and confidence are uploaded to the firestore cloud server in real-time; 3) A JAVA code was written to query the cloud server and get the latest uploaded observations from the surrendering users within a specific spatial range (i.e., function in the coverage range of the utilized signal source for inter-user measurements). This code was included in the developed real-time JAVA code for non-collaborative localization; 4) When the BLE broadcasting option is activated by the user, the user can query the existing data on the server to retrieve the required data in real-time; and 5) by retrieving the required data, the RT-MUCIP algorithm was implemented to estimate collaborative positioning solution. A JAVA code was written to estimate the collaborative localization solution using the proposed collaborative localization approach.

2. Test Field and Setups

The experimental environment is a laboratory with a large open-space area; see Figure 2 First, the Wi-Fi signatures for distributed RPs were recorded statically to set up the test field using the developed application (red rectangular marker in Figure 2). Distributed test points were marked to evaluate the estimated position, the yellow triangle markers in Figure 2. Five smartphones with different models were utilized as agent nodes and were labeled (A, B, C, D, and E). Three tests were conducted to evaluate the proposed method's performance in different scenarios. The following sections present the performance of the conducted tests.

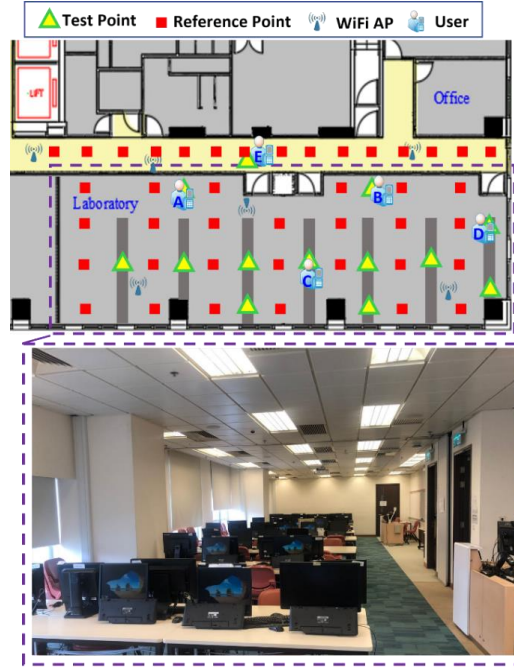


Figure 2 Test Area

3. Results and Evaluations

a) Test 1: Group of Stationary Users

The first scenario involves a group of stationary users placed in locations designed to provide different density levels of users. A user's position and confidence were first estimated individually based on WiFi RSS-based fingerprinting. Subsequently, inter-user BLE RSS measurements were exchanged among the collaborated users. The inter-user distance was estimated based on the path loss propagation model and then weighted. Table 2 lists the positioning results for the non-collaborative and collaborative approaches and the percentage of the improved accuracy. It is worth noting that the agent node D achieved the most remarkable improvement in positioning accuracy, reaching 16%. Node D, the node with the lowest confidence, achieved improvements in both position estimation and confidence. Additionally, the confidence level of the estimated collaborated solution is improved for all nodes. The average position accuracy improvement reached ~12.5%.

Table 2 Results for Test (1) Group of 5 Stationary users. Four users were inside a large room, and one was in the corridor (outside the room). The standalone solution was estimated using Wi-Fi fingerprinting. The locations of the fixed Wi-Fi APs did not involve in collaborative solution. The user D was placed far away from the other users to examine the user density.

User ID	Standalone user solution		Collaborative solution		Position improving (%)
	Expected Error (confidence) (m)	Error (m)	Solution precision (From variance) (m)	Error (m)	
A	2.1	3.1	1.82	2.75	11%
B	1.82	3.43	2.06	2.95	14%
C	2.25	3.65	1.86	3.32	9%
D (low user density)	3.3	3.15	1.78	2.64	16%
E	1.35	1.70	1.35	1.70	-

b) Test 2: Involving the Fixed Wi-Fi APs as Anchor Nodes

This test was conducted similarly to the first, except that Wi-Fi APs fixed at specific locations were incorporated into collaboration localization. The locations of these APs were measured, and a higher confidence level was assigned to them. It is important to note that the location and propagation parameters of the fixed APs can be estimated using radio map fingerprints, as explained in detail

in (Mansour et al., 2023). During this study, we chose to use measured AP locations instead of estimating them to avoid potential errors. Table 3 lists the results of the five users in non-collaborative and collaborative approaches. The collaborative localization results were enhanced after incorporating the Wi-Fi APs as ANs. Agent node D achieved the highest positioning accuracy improving with enhancement reaching 28%. Moreover, the average position accuracy improvement reached ~20.5%. In tests 1 and 2, the standalone solution of node E reached high positioning accuracy and confidence. Thus, in the collaborative approach, node E was not considered a target node to improve its position; it only participated in improving the performance of the surrounding nodes.

Table 3 Results of Test (2)

User ID	Standalone user solution		Collaborative solution		Position improving (%)
	Expected Error (confidence) (m)	Error (m)	Solution precision (From variance) (m)	Error (m)	
A	2.1	3.1	1.25	2.54	18%
B	1.82	3.43	1.79	2.72	20%
C	2.25	3.65	1.68	3.05	16%
D (low user density)	3.3	3.15	1.52	2.25	28%
E	1.35	1.70	1.35	1.70	-

c) Test 3: Group of Walked and Stationary Nodes

This test involved collaboration among a group of stationary (A, C, and D) and walking users (B and E) that walked between distributed test points. This test was specifically designed to offer stationary nodes the opportunity to qualify as temporary anchor nodes and enhance the positioning accuracy of the walking users. The position and confidence of walked users were estimated individually by fusing pedestrian dead reckoning (PDR) and WiFi fingerprinting solutions using an extended Kalman filter. The position of each walked node was recorded when crossing a test point to evaluate the estimated individual and collaborative solutions. Table 4 lists the results obtained for the two walked agent nodes in non-collaborative and collaborative approaches. The error at each test point crossed by the walked node was estimated; the error listed in Table 4 is the mean error estimated from the errors recorded for all crossed testing points. Users A and D were successfully upgraded to TAN during this test, which improved the positioning accuracy for nodes B and E. Specifically, the position of node B experienced an improvement of 11%, while node E achieved an enhancement of 18%. These results demonstrate the effectiveness of utilizing qualified stationary nodes as TANs to enhance the positioning performance of other nodes in the collaborative localization system.

Table 4 Results of Test (3).

User ID	Non-collaborative	Collaborative	Position improving (%)
	Error (m)	Error (m)	
B	4.36	3.85	11%
E	3.85	3.15	18%

V. CONCLUSIONS

In the current era, indoor positioning has become crucial for several applications, ranging from lifesaving to advertising. Thus, extending positioning availability to indoor areas and estimating indoor positioning solutions with high confidence is highly required. During the past two decades, several standalone user-based indoor positioning systems (SU-IPSs) have been developed; however, these systems are highly prone to low position confidence or even position unviability in challenging indoor positioning scenarios. Thereby, the concept of real-time multi-user collaborative indoor localization has emerged to boost SU-IPSs in these scenarios by exchanging implicit or explicit inter-user measurements. In the literature, different collaborative indoor localization approaches have been proposed. However, this research goes beyond existing studies by leveraging multi-user collaboration in diverse indoor scenarios, adapting the collaborative localization approach to accommodate varying user densities, and identifying stationary users with high position confidence to serve as fixed anchor nodes, thereby augmenting the anchor node density. Several tests were conducted in three real-world scenarios to evaluate the performance of the proposed scheme. The proposed method achieved improving position accuracy by ~12.5% compared to SU-IPS solution, for users in stationary mode, and of ~16% for user in area with sparse users. Incorporating fixed Wi-Fi APs as anchor nodes enhanced the collaborative results by ~20.5%. When stationary users upgraded to temporary anchor nodes, the position estimation of walking users improved by ~15%. The overall results lead us

to conclude that, without the need to use external resources, collaborative localization can improve the localization accuracy of the collaborated nodes, compared to the non-collaborative solution.

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